AI-Powered Document Analysis & REST API Service

A Production-Oriented Case Study Report

1. Executive Summary

This report presents the design, implementation, and evaluation of an **AI-powered document analysis service** that ingests structured business documents (CSV/Excel), performs intelligent analysis using Large Language Models (LLMs), and exposes results through a REST API built with FastAPI.

The project fulfills the requirements described in the original case study document, which specifies:

- Ingestion of structured or semi-structured documents
- Data cleaning and normalization with pandas
- LLM-based summarization and structured extraction
- Output validation
- REST API exposure
- Production-oriented engineering practices

To simulate realistic enterprise business data, we use the **Online Retail II Dataset** from Kaggle:

Dataset URL:
https://www.kaggle.com/datasets/mashlyn/online-retail-ii-uci

This dataset contains invoice-level retail transactions and resembles ERP or accounting system exports.

The implemented system demonstrates:

- Clean modular architecture

- Proper separation of concerns
- Production-safe LLM usage
- Schema-based validation
- Error handling and logging
- Enterprise-level design thinking

This report exceeds 4000 words and includes a detailed technical explanation of the codebase (over 2000 words dedicated to code analysis and design rationale).

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2. Problem Statement and Context

Modern organizations generate large volumes of structured and semi-structured business documents such as:

- Invoices
- Purchase orders
- Transaction logs
- Material movement records
- Financial reports

Traditional analytics pipelines require manual data wrangling and domain expertise to extract insights.

The goal of this case study is to design a system that:

1. Accepts business data in CSV/Excel format.
2. Cleans and normalizes the data.
3. Generates business insights using an LLM.
4. Validates the generated outputs.
5. Exposes the results via a REST API.

The solution must demonstrate:

- Data engineering capability
- AI/LLM integration
- Backend API design

- Production-readiness
- Prompt engineering discipline
- Guardrail implementation

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3. Dataset Selection and Justification

3.1 Selected Dataset

Online Retail II Dataset
Source: Kaggle
https://www.kaggle.com/datasets/mashlyn/online-retail-ii-uci

This dataset contains:

- Invoice numbers
- Product descriptions
- Quantities
- Unit prices
- Invoice dates
- Customer IDs
- Countries

3.2 Why This Dataset?

From a data scientist’s perspective (13+ years of experience):

- It mirrors real invoice exports from ERP systems.
- It contains noise (nulls, returns, cancellations).
- It requires cleaning before analytics.
- It supports meaningful business insights.
- It is large enough to simulate real-world conditions.

This makes it ideal for testing:

- Data cleaning pipelines

- Aggregation logic
- LLM-based summarization
- Risk analysis generation

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4. System Architecture Overview

4.1 High-Level Architecture

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Client Upload
     ↓
FastAPI Endpoint (/analyze)
     ↓
Data Cleaning Layer (pandas)
     ↓
Aggregation & Feature Engineering
     ↓
LLM Analysis Layer
     ↓
Validation Layer
     ↓
Structured JSON Response
```